

BIRLA INSTITUTE OF TECHNOLOGY MESRA, RANCHI
DEPARTMENT OF MATHEMATICS

MA24103 Mathematics-II

Session: SP 2026

Tutorial - 3 (Module III)

1. Find the Fourier series of $f(x) = \pi^2 - x^2$ in $-\pi < x < \pi$.
2. Find the Fourier series for the function $f(x)$ if $f(x)$ is defined in $-\pi < x < \pi$ as

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ \sin x & 0 < x < \pi \end{cases}$$

Hence deduce that

$$\frac{1}{4}(\pi - 2) = \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots$$

3. Obtain a Fourier expression for $\sqrt{1 - \cos x}$ in the interval $-\pi < x < \pi$.
4. Find the Fourier series for the function

$$f(x) = \begin{cases} -h & -\pi < x < 0 \\ h & 0 < x < \pi \end{cases}$$

here, $h \neq 0$. Hence deduce that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

5. Consider the function $f(x) = e^x$ in $0 < x < 2$.
 1. Determine the Fourier series of $f(x)$.
 2. Determine the half-range sine series of $f(x)$.
 3. Determine the half-range cosine series of $f(x)$.
6. Determine the half-range Fourier sine series for $f(x) = x(\pi - x)$ in $0 < x < \pi$. Hence deduce

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^3} = \frac{\pi^3}{32}$$

7. Determine the half-range sine and cosine series for $f(x) = \sin(x)$, where $0 < x < \pi$.
8. Determine the half-range Fourier cosine series for $f(x) = \sin\left(\frac{\pi x}{l}\right)$, where $0 < x < l$.
9. Obtain the half range Fourier cosine and sine series for

$$\begin{cases} 0 & 0 < x < \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} < x < \pi \end{cases}$$

10. Solve the partial differential equations using method of separation of variables

a) $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$.

b) $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$, given that $u(0, y) = 8e^{-3y}$.

c) $4 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$, $u(0, y) = 3e^{-y} - e^{-5y}$.

d) $3 \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0$, given that $u(x, 0) = 4e^{-x}$.

11. Find the temperature in a bar of length 2 whose ends kept at zero and lateral surface insulated if the initial temperature is $\sin \frac{\pi x}{2} + 3 \sin \frac{5\pi x}{2}$.
12. A tightly stretched string with fixed end points $x = 0$ and $x = \pi$ is initially at rest in its equilibrium position. If it is set vibrating by giving to each of its points an initial velocity $\left(\frac{\partial y}{\partial t}\right)_{t=0} = 0.03 \sin x - 0.04 \sin 3x$ then find the displacement $y(x, t)$ at any point of string at any time t .
13. The temperature distribution in a bar of length π , which is perfectly insulated at the ends $x = 0$ and $x = \pi$ is governed by the partial differential equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$. Assuming the initial temperature as $u(x, 0) = f(x) = \cos 2x$, find the temperature distribution at any instant of time.
14. Solve the equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$, representing the vibration of a string of length l , fixed at both ends, given that $y(0, t) = 0, y(l, t) = 0, y(x, 0) = f(x)$ and $\frac{\partial y}{\partial t}(x, 0) = 0, 0 < x < l$.
15. Solve the equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, representing the vibration of a string of length l , fixed at both ends, given that $u(0, t) = 0, u(l, t) = 0, u(x, 0) = 0$ and $\frac{\partial u}{\partial t}(x, 0) = \cos x, 0 < x < l$.
16. Solve the equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, representing the vibration of a string of length l , fixed at both ends, given that $u(0, t) = 0, u(l, t) = 0, u(x, 0) = 0$ and $\frac{\partial u}{\partial t}(x, 0) = \begin{cases} x, & \text{if } 0 < x < l/2 \\ l - x, & \text{if } l/2 < x < l. \end{cases}$

***** End *****